SINKGRAB · RISK REGISTER

SinkGrab risk register, hazard analysis and design FMEA

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SKG-DEC-001 O2, O4, O5): R1 and R9 restated, R10 added; H5 controls; FMEA F1, F3 and F11 rewritten for the load indicator and the crank slip clutch
0.1	2026-10-24	Amish Chadha	First risk register with preliminary hazard analysis and a first-pass design FMEA (kit 1.9.1 upgrade, stage P3)

SinkGrab risk register, hazard analysis and design FMEA

The top project risk is that the caisson rings stop sinking before 3 m because skin friction is higher than the 1.66 kPa self-weight can overcome (R1); since Amish's decision O2 of 2026-10-25 a bentonite slurry in the overcut annulus is part of the method, and the test pit shows whether it is enough. The narrow slip window of the crank slip clutch chosen under O4 follows (R9). The top hazards are a person falling into or entering the shaft and a load or ring dropping, all controlled by keeping everyone at the surface behind a barrier with the doors shut. The design FMEA's highest items are the slip clutch tightened past its setting and the load indicator's tell-tale not read (RPN 30 and 27), both kept in check by the CalRig setting, the paint-marked locknut and the reading at every lift; the crosshead and the scraper fold follow, each designed against its failure (SKG-DEC-001 K8, K6). Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high; for detection, 5 is hardest to detect).

> **Safety:** SinkGrab is lifting equipment worked over an open well shaft, with a hand-cranked chain drive and loads hanging over the well. This register is a paper assessment for a TRL 3 design; it does not show that any part is safe.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R1	Rings stop sinking short of 3 m	Skin friction above 1.66 kPa; the central unaided estimate is 2 kPa and soil closes on the ring above the overcut (independent review)	PR-001 not met; well not deepened enough	4	5	Design lead	Bentonite slurry fed into the overcut annulus as part of the method, decided 2026-10-25 (O2); tried in the test pit before a partner well; jacking against a reaction is the next step if the test pit shows it is not enough
R2	Grab fills poorly in denser sand or laterite	Closing force margin about 30 % in loose sand (STEM S2)	Bite below 20 L (PR-003); slower deepening	3	3	Design lead	Test-pit bite trial in each soil (PR-013) before a partner well
R3	Cycle over 3 min	Hoist limited by human power (78 s of 3.19 min)	PR-004 not met; more days per well	4	2	Design lead	Timed trial decides (decision 27A); accept if the daily output suffices
R4	No co-design partner	Neither candidate approached yet	No trial well, no validation of user requirements	3	4	Amish (owner)	Approach the Kerala Ground Water Department, then the Ethiopian NGO (SKG-DDR-001 item 12)
R5	HatchSide tripod not available or changed	Shared block, rated for material handling only	Interfaces (head sheave, foot cradle) no longer fit	2	4	Design lead	Keep the interface drawings current; any HatchSide change is checked against SinkGrab
R6	Fabricators lack plate rolls or a profile cutter	Village workshops	Shells and profile-cut parts made in a city	3	2	Co-design partner	Faceted skin fallback (SKG-DEC-001 value engineering); profile-cut parts ordered as a set
R7	Concept choice changes after the build plan	Options A and B within 10 % (O1)	Redrawing and rebuilding the plan	2	3	Amish (owner)	Decide O1 before any TRL 4 build
R8	Under-curb scraper claims in Chinese caisson patents	Targeted search not yet run	Release blocked or redesign	2	3	Amish (owner)	Search before public release (SKG-DDR-001 item 14)
R9	Narrow slip window for the crank slip clutch	Steady full-grab pull 1,068 N against a lowest slip of 1,127 N on the third rope	The clutch slips on a steady hoist and work	3	4	Design lead	Set and checked on CalRig before each campaign on every layer; facings kept dry inside

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
		layer (6 %); facing friction moves with moisture and glazing	stops, or it slips high				the guard; load indicator read every lift; an uprated tripod (O4 option B) if the trials show more bite is worth it
R10	Four people needed to turn the scraper	Decision O5: the 2.0 m T-bar does not clear the tripod legs	Crew short on the day; scraping slower or pushed past 100 N each	2	2	Site lead	Crew of five or six planned for scraping days; briefing card says four at the T-bar

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Fall into the shaft	Open well mouth while the grab passes	Death or serious injury	5	2	Doors closed whenever the grab is above them; barrier 2 m outside the tripod feet; nobody on the doors during a lift (BLD stop 3)	1
H2	Entry into the well	Freeing a snag, retrieving a tool	Asphyxiation, drowning, struck by load	5	2	Nobody enters while SinkGrab works; recovery line on the grab; entry only as a separate operation with gas test, harness and rescue (BLD stop 8)	1
H3	Bad air at the well mouth	Carbon dioxide, hydrogen sulphide or low oxygen in the shaft	Collapse of a person leaning over the shaft	5	2	Four-gas detector and carbon dioxide monitor lowered and read at three depths before work, after breaks and before leaning over (PR-012, K5, K22)	1
H4	Grab or scraper falls into the well or onto people	Rope, splice, pin or shackle failure	Death or serious injury below or beside the line	5	2	Line factor 10.2; proof load 196 kg before use; nobody under or beside the line; daily inspection (BLD stops 2, 3, 9)	2
H5	Tripod overloaded	Snag while hoisting	Tripod collapse onto the crew	5	2	Crank slip clutch caps the capstan pull at the 150 kg rating, set on CalRig (O4, 2026-10-25); the load indicator shows the peak of each lift; proof at 1.25 x the rated load, 188 kg, inside the 225 kg proof	2
H6	Load runs back while lowering	Brake not set; cranks on while lowering	Spinning cranks strike the crew; grab drops	4	2	Weighted band brake on when let go, geometry corrected (K2); cranks off before lowering; pawl (BLD stop 4)	1
H7	Fingers or clothing in the chain or brake	Guard off; reaching in	Crush and cut injuries	3	2	Chain guard on M6 screws into clip angles; never turn the cranks with it off	1
H8	Rings drop or tilt suddenly	Uneven undercutting	Ring damage; lining out of line	3	3	Sump at most 330 mm below the edge; tilt read every cycle; stop	2

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						undercutting at 1 in 80 (BLD stop 7)	
H9	Back injury from heavy pieces	Lifting the grab, a full tub or the capstan by hand	Musculoskeletal injury	3	3	Pieces of 25 kg or less carried by two; grab only on the line (K3); tub slid, not lifted (K4)	1
H10	Contaminated water after the work	Sediment, tools and hands in the well	Illness of the households who use it	4	2	Clean tools; disinfect the well after the work (Oxfam guidance; SKG-REG-001 G12)	2
H11	Three people at the capstan during the hoist	Third person on the long handle	Struck by a handle or the line	3	2	All three let go together on the stop call; nobody between the capstan and the lead sheave (BLD stop 4)	1
H12	Jerk crew beside the loaded line (option B only)	Spudding at the well mouth	Struck by the line or bar if the bailer jams	3	3	Applies only if Amish chooses option B (O1)	3

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Dead-end load indicator	Show the highest pull of each lift	Tell-tale lost or stuck; springs set or corroded	Peak not seen, so a clutch slipping high goes unnoticed	Mud on the rod; O-ring perished; spring stack rusted	3	3	3	27	Read and reset at the doors after every lift; stainless rod; spare O-rings on the frame; recalibrated on CalRig with the clutch
F2	Band brake	Hold the drum when the lever is let go	Weight slackens instead of sets; lining wet or worn	Load runs back	Link pin on the wrong side of the pivot (kit 1.7 model); lining glazing	5	2	2	20	Lever geometry corrected (K2); brake check before each lift (first checks); inspect lining daily
F3	Crank slip clutch	Cap the capstan pull at the 150 kg rating	Slips high: locknut tightened past its setting or facings rusted to the discs	Line can pass the tripod's rating	Crew tightens it to stop slipping; clutch left wet and standing	5	2	3	30	Locknut paint-marked and held by its tab washer; briefing card and stop 6 say "never tighten the clutch"; slip checked against a fixed stop at the start of each day; load indicator read every lift; CalRig check before each campaign
F4	Scraper fold	Fold the arms to pass the ring	Arms stay open	Head jams under the ring; nobody may enter to free it	Sleeve not sliding; cross pin missing; arm eye on the hub	4	2	3	24	Pivots at 60 mm so the eyes clear the hub (K6); fold check on the ground through a 0.8 m ring; cross pin checked at every pole change
F5	Closing line	Carry the grab	Abrasion or cut	Grab lost in the well	Rubbing on the door notch or a sheave flange; UV	4	2	2	16	Daily inspection; replace at the first damaged strand; fairlead only in option B
F6	Crosshead on the hinge pin	Keep the tackle in line	Crosshead slides along the pin	Line rubs the head; uneven closing	Nothing held it axially (kit 1.7 model)	3	3	3	27	Two set collars (K8); check on assembly
F7	Drum bearing bolts	Hold the drum bearings to the pads	Bolts cannot be fitted or work loose	Drum moves; chain jumps	Nuts would sit inside the pad posts (kit 1.7 layout)	3	2	2	12	Captive bolts pushed up before the pads are welded (K10); check torque daily
F8	Crank handles	Turn the cranks	Handle hardware	Guard damaged;	Bolt and nut on the inner	3	2	2	12	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
			strikes the chain guard	crank jams mid-hoist	face of the arm					Welded stub axles, nothing on the inner face (K11)
F9	Doors	Cover the shaft, carry the tub	Door lifts or a tub tips in	Fall or spoil into the well	Door not seated; tub on the meeting edge	4	2	2	16	Doors close flat on the vertical legs; tub stands on the mesh, slid off on a board (K4)
F10	Ballast saddles	Add weight on the high side	Saddle drops into the annulus	Saddle lost; ring knocked	Line not made off	2	2	2	8	Each saddle on its own made-off line
F11	Crank slip clutch	Carry the hoisting torque up to its setting	Slips low: facings wet or oily	Hoist stops with the load on the pawl	Rain or grease inside the guard	2	3	2	12	Guard closed with a drain hole; no grease near the clutch; facings dried or changed; replaces the crank pin's fatigue mode (STEM S6)

Highest RPN: F3 (crank slip clutch tightened past its setting, 30), then F1 (load indicator not read, 27) and F6 (crosshead, 27). The responses are partly procedural, so they stay on the briefing card and in the first checks; the crank shear pin's modes (RPN 45 and 36) are gone with the pin.